

IsPilot API - Work Checkpoint

Status: ☒ Production validation passed; API contract is stable; next gate is business-scenario validation and channel integration

Date: 2026-08-29

Production Endpoint: <https://ispilot-api-46y2f3tyja-uc.a.run.app>

Current verified state

The API is live and operational. The production service was validated with a real bearer-token request, and the root cause of the earlier failure was a malformed payload / quoting issue rather than a broken deployment.

Verified facts:

- Cloud Run API is live and responding
- OAuth bearer token authentication works with the correct service-account impersonation flow
- Valid JSON request payloads are accepted by the API
- A session reuse pattern works when a valid `session_id` is supplied
- No redeploy was required to restore service health

Working smoke-test pattern

```
$token = gcloud auth print-identity-token --impersonate-service-account=sa-tot-osa@corp-stro-salesinventory-prod.iam.gserviceaccount.com
$body = '{"user_id":"debug-user","message":"Give me a simple answer about inventory","session_id":"5411387247947677696"}'

curl.exe -sS -X POST "https://ispilot-api-46y2f3tyja-uc.a.run.app/chat" `
  -H "Content-Type: application/json" `
  -H "Authorization: Bearer $token" `
  --data-raw $body
```

This is the canonical production validation command to keep for future checks.

Completed work

API runtime and contract

- FastAPI chat endpoint is live
- Request model accepts `user_id`, `message`, and optional `session_id`
- Auth accepts valid bearer token and local API-key flow for dev/testing
- Session creation/reuse is implemented
- Response contract is standardized

Observability and hardening

- OpenAPI docs are available and documented
- Request/response logging is in place
- Audit logging is active
- Metrics and latency tracking were integrated
- Firestore fallback is present for resilience

Production documentation

- The live smoke test is documented in the API README
 - Auth guidance and payload requirements are recorded for reuse
 - Root-level guidance clarifies the split between business logic and API runtime layer
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Remaining work

Business-scenario validation

This is the next meaningful quality gate. The API contract is stable, but we still need to validate end-to-end business quality for realistic questions such as:

- daily summary checks
- store performance questions
- rankings and comparisons
- recommendation quality
- multi-turn session behavior

Channel integration

Teams / Copilot should be treated as an application layer on top of this API, not as the primary debugging surface. The sequence should be:

1. API contract stable
2. business validation passed
3. enterprise channel integration enabled

Regression guardrails

The passwordless bearer-token smoke test should remain as the default operational regression step after future deployment or auth changes.

Documentation integration

Date: 2026-08-30

All deployment instructions have been integrated into README.md. The separate DEPLOYMENT.md file is no longer needed. The repository now follows a cleaner structure:

Files kept as separate reference documents:

- [README.md](#) - Complete API documentation including deployment, architecture, authentication, and local development
- [WORK_CHECKPOINT.md](#) - Ongoing work status and task tracking
- [docs/PERMISSIONS_AND_IAM.md](#) - IAM setup and security role configuration
- [docs/SESSION_SERVICE_FIRESTORE_FALLBACK.md](#) - Session management implementation
- [docs/AUTHENTICATION_CHANGES.md](#) - Authentication evolution and details

This consolidation makes the documentation more navigable and reduces file fragmentation.

Current status summary

Status: Production API is validated and stable.

Next gate: business-scenario validation and downstream Teams/Copilot integration.

No immediate redeploy needed: the earlier issue was a malformed request, not a service outage.

Planned next steps

1. Validate real business questions across the agent stack
2. Confirm answer quality and routing decisions with retail scenarios
3. Finalize the API contract for enterprise consumers
4. Wire Teams / Copilot as a channel on top of the validated API

This is the correct operational sequence: validate the API contract, validate the business answers, then expose it to end-user channels.

Steps:

1. Security review (OAuth tokens, secret rotation, IAM least privilege)
2. Load testing (concurrent requests, sustained throughput)
3. Chaos testing (Vertex AI timeouts, Firestore unavailable, auth failures)
4. Production deployment procedures
5. Health check tuning and auto-healing
6. Rollback procedures and disaster recovery
7. Incident response playbook

Estimated Checkpoints: 10-12

Sprint 5: Advanced Observability

Goal: Enhanced visibility into system behavior

Steps:

1. Distributed tracing (Cloud Trace integration)
 2. Custom dashboards (Cloud Monitoring)
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3. SLO-based alerts and tracking
4. Error budget calculations
5. Weekly metrics reports automation
6. Performance profiling and optimization recommendations

Estimated Checkpoints: 8-10

Sprint 6: Feature Enhancements (Future)

Goal: New capabilities based on operational insights

Steps (To be defined):

- Multi-turn conversation optimization
- Session persistence improvements
- Custom model endpoints
- Rate limiting policies
- Advanced caching strategies

Status: Pending user requirements

Next Steps (If Continuing)

Immediate Actions (When Resuming)

1. Review WORK_CHECKPOINT.md this section to understand remaining work
2. Choose next sprint based on priority (Metrics Integration → Production Hardening → Advanced Observability)
3. Check git log for latest commit: `git log --oneline -1`
4. Verify all code is current: `git status`
5. Set up environment: `python -m venv venv && source venv/bin/activate`
6. Install dependencies: `pip install -r requirements.txt`

For Sprint 3 Start

1. Review metrics definitions in `app/utils/metrics_definitions.py`
2. Integrate decorators in `app/services/vertex_agent_client.py`
3. Integrate context managers in `app/services/session_service.py`
4. Update DEPLOYMENT.md with staging environment steps
5. Deploy and validate metrics

Summary

Sprint 2 is **COMPLETE** with all 7 steps implemented, tested, and validated. The IsPilot API now has:

- ☒ Comprehensive OpenAPI documentation
- ☒ Enhanced observability with detailed logging
- ☒ Cloud Monitoring metrics ready for deployment

- ☒ 31/31 validation checkpoints passing
- ☒ Zero regressions from prior phases
- ☒ Production-ready error handling and graceful degradation

All changes committed and ready for staging/production deployment.